

Mouse auditory experiment

A first plan

October 8, 2026

We want to play short tones to a mouse and record how its brain responds, using SONIC as a starting point.

1. Choose the sounds

- Choose the frequency, duration, loudness, and gap between tones.
- Add short fades at both ends to avoid clicks.
- Save the sequence and settings so we can identify every tone.
- SONIC used 10 ms tones with 2 ms fades in awake sheep. Our mouse settings and awake versus anesthetized state remain open.

2. Set up the speaker

- Check what audio equipment the lab already has.
- Consider the TDT MF1 speaker and SA1 amplifier used in SONIC.
- Keep the speaker in a fixed position.
- Measure each tone's loudness with a suitable calibrated microphone at the ear position.

3. Keep everything on one timeline

- Schedule each tone with an electrical timing pulse.
- Record the pulse alongside the neural signals.
- Synchronize the microphone recording to check when the sound actually reaches the ear.
- If we add behavior video, synchronize its frame times too.

Start here

- Identify the neural recorder and available audio equipment.
- Test the sound, timing, and unwanted electrical pickup on the bench.
- Once those checks work, use the team's mouse protocol to start with tones separated by silence.

References: SONIC, bioRxiv v1, Methods 6.1–6.2 (p.11) and 6.6 (p.13); TDT speaker information; TDT timing information. This is our proposed mouse setup.